

instruction register interrupt interrupt-driven I/O I/O module locality main memory memory hierarchy	multicore multiprocessor processor program counter programmed I/O register secondary memory	spatial locality stack stack frame stack pointer system bus temporal locality
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Review Questions

- 1.1. List and briefly define the four main elements of a computer.
- 1.2. Define the two main categories of processor registers.
- 1.3. In general terms, what are the four distinct actions that a machine instruction can specify?
- 1.4. What is an interrupt?
- 1.5. How are multiple interrupts dealt with?
- 1.6. What characteristics distinguish the various elements of a memory hierarchy?
- 1.7. What is cache memory?
- 1.8. What is the difference between a multiprocessor and a multicore system?
- 1.9. What is the distinction between spatial locality and temporal locality?
- 1.10. In general, what are the strategies for exploiting spatial locality and temporal locality?

Problems

- 1.1. Suppose the hypothetical processor of Figure 1.3 also has two I/O instructions:

0011 = Load AC from I/O

0111 = Store AC to I/O

In these cases, the 12-bit address identifies a particular external device. Show the program execution (using format of Figure 1.4) for the following program:

1. Load AC from device 5.
2. Add contents of memory location 940.
3. Store AC to device 6.

Assume that the next value retrieved from device 5 is 3 and that location 940 contains a value of 2.

- 1.2. The program execution of Figure 1.4 is described in the text using six steps. Expand this description to show the use of the MAR and MBR.
- 1.3. Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields. The first byte contains the opcode and the remainder an immediate operand or an operand address.
 - a. What is the maximum directly addressable memory capacity (in bytes)?
 - b. Discuss the impact on the system speed if the microprocessor bus has
 1. a 32-bit local address bus and a 16-bit local data bus, or
 2. a 16-bit local address bus and a 16-bit local data bus.
 - c. How many bits are needed for the program counter and the instruction register?
- 1.4. Consider a hypothetical microprocessor generating a 16-bit address (e.g., assume that the program counter and the address registers are 16 bits wide) and having a 16-bit data bus.
 - a. What is the maximum memory address space that the processor can access directly if it is connected to a “16-bit memory”?

- b. What is the maximum memory address space that the processor can access directly if it is connected to an “8-bit memory”?
 - c. What architectural features will allow this microprocessor to access a separate “I/O space”?
 - d. If an input and an output instruction can specify an 8-bit I/O port number, how many 8-bit I/O ports can the microprocessor support? How many 16-bit I/O ports? Explain.
- 1.5** Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8-MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate across the bus that this microprocessor can sustain in bytes/s? To increase its performance, would it be better to make its external data bus 32 bits or to double the external clock frequency supplied to the microprocessor? State any other assumptions you make and explain. *Hint:* Determine the number of bytes that can be transferred per bus cycle.
- 1.6** Consider a computer system that contains an I/O module controlling a simple keyboard/printer Teletype. The following registers are contained in the CPU and connected directly to the system bus:
- INPR: Input Register, 8 bits
 - OUTR: Output Register, 8 bits
 - FGI: Input Flag, 1 bit
 - FGO: Output Flag, 1 bit
 - IEN: Interrupt Enable, 1 bit

Keystroke input from the Teletype and output to the printer are controlled by the I/O module. The Teletype is able to encode an alphanumeric symbol to an 8-bit word and decode an 8-bit word into an alphanumeric symbol. The Input flag is set when an 8-bit word enters the input register from the Teletype. The Output flag is set when a word is printed.

- a. Describe how the CPU, using the first four registers listed in this problem, can achieve I/O with the Teletype.
 - b. Describe how the function can be performed more efficiently by also employing IEN.
- 1.7** In virtually all systems that include DMA modules, DMA access to main memory is given higher priority than processor access to main memory. Why?
- 1.8** A DMA module is transferring characters to main memory from an external device transmitting at 9600 bits per second (bps). The processor can fetch instructions at the rate of 1 million instructions per second. By how much will the processor be slowed down due to the DMA activity?
- 1.9** A computer consists of a CPU and an I/O device D connected to main memory M via a shared bus with a data bus width of one word. The CPU can execute a maximum of 106 instructions per second. An average instruction requires five processor cycles, three of which use the memory bus. A memory read or write operation uses one processor cycle. Suppose that the CPU is continuously executing “background” programs that require 95% of its instruction execution rate but not any I/O instructions. Assume that one processor cycle equals one bus cycle. Now suppose that very large blocks of data are to be transferred between M and D .
- a. If programmed I/O is used and each one-word I/O transfer requires the CPU to execute two instructions, estimate the maximum I/O data transfer rate, in words per second, possible through D .
 - b. Estimate the same rate if DMA transfer is used.
- 1.10** Consider the following code:
- ```

for (i = 0; i < 20; i++)
 for (j = 0; j < 10; j++)
 a[i] = a[i] * j

```
- a. Give one example of the spatial locality in the code.
  - b. Give one example of the temporal locality in the code.

**1.11** Generalize Equations (1.1) and (1.2) in Appendix 1A to  $n$ -level memory hierarchies.

**1.12** Consider a memory system with the following parameters:

$$\begin{aligned} T_c &= 100 \text{ ns} & C_c &= 0.01 \text{ cents/bit} \\ T_m &= 1,200 \text{ ns} & C_m &= 0.001 \text{ cents/bit} \end{aligned}$$

- What is the cost of 1 MByte of main memory?
- What is the cost of 1 MByte of main memory using cache memory technology?
- If the effective access time is 10% greater than the cache access time, what is the hit ratio  $H$ ?

**1.13** A computer has a cache, main memory, and a disk used for virtual memory. If a referenced word is in the cache, 20 ns are required to access it. If it is in main memory but not in the cache, 60 ns are needed to load it into the cache (this includes the time to originally check the cache), and then the reference is started again. If the word is not in main memory, 12 ms are required to fetch the word from disk, followed by 60 ns to copy it to the cache, and then the reference is started again. The cache hit ratio is 0.9 and the main-memory hit ratio is 0.6. What is the average time in ns required to access a referenced word on this system?

**1.14** Suppose a stack is to be used by the processor to manage procedure calls and returns. Can the program counter be eliminated by using the top of the stack as a program counter?

## APPENDIX 1A PERFORMANCE CHARACTERISTICS OF TWO-LEVEL MEMORIES

In this chapter, reference is made to a cache that acts as a buffer between main memory and processor, creating a two-level internal memory. This two-level architecture exploits a property known as locality to provide improved performance over a comparable one-level memory.

The main memory cache mechanism is part of the computer architecture, implemented in hardware and typically invisible to the OS. Accordingly, this mechanism is not pursued in this book. However, there are two other instances of a two-level memory approach that also exploit the property of locality and that are, at least partially, implemented in the OS: virtual memory and the disk cache (Table 1.2). These two topics are explored in Chapters 8 and 11, respectively. In this appendix, we look at some of the performance characteristics of two-level memories that are common to all three approaches.

**Table 1.2** Characteristics of Two-Level Memories

|                                     | Main Memory Cache               | Virtual Memory (Paging)                     | Disk Cache       |
|-------------------------------------|---------------------------------|---------------------------------------------|------------------|
| Typical access time ratios          | 5 : 1                           | $10^6$ : 1                                  | $10^6$ : 1       |
| Memory management system            | Implemented by special hardware | Combination of hardware and system software | System software  |
| Typical block size                  | 4 to 128 bytes                  | 64 to 4096 bytes                            | 64 to 4096 bytes |
| Access of processor to second level | Direct access                   | Indirect access                             | Indirect access  |